NOTRE DAME UNIVERSITY BANGLADESH

Machine Learning Lab Report-02

Course Code: CSE4214

Course Title: Machine Learning Lab

Lab Task Topic: Linear & Multiple Linear Regression

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1 Objective

The objective of this Machine Learning lab is to understand and implement regression techniques using real-world datasets. This lab focuses on applying Simple Linear Regression and Multiple Linear Regression to analyze the relationship between independent variables and a target variable. The experiments aim to develop predictive models, interpret feature influence, and evaluate model performance using appropriate regression metrics.

2 Dataset Description

Three different datasets are used in this lab to demonstrate the application of regression algorithms.

Dhaka Home Prices Dataset

The `dhaka homeprices.csv` dataset is used for implementing Simple Linear Regression. This dataset contains housing-related information from Dhaka city, where a single independent feature is used to predict house prices. The dataset helps illustrate how a dependent variable varies linearly with one explanatory variable.

Insurance Dataset

The `insurance.csv` dataset is used for implementing Multiple Linear Regression. It contains multiple attributes related to insurance policyholders. The dataset is used to analyze how multiple independent variables collectively influence insurance-related outcomes.

Car Data Dataset

The `cardata.csv` dataset is also used for Multiple Linear Regression. It includes attributes such as `speed`, `car_age`, `experience`, and `risk`. These features are used to predict a target variable by learning the combined effect of multiple factors on the outcome.

3 Linear Regression Implementation

4 KNN as Classifier [As Regressor in another program]

[1]:
```
#Assigning Feature and local variables
#First Feature
weather = ['Sunny','Sunny',
'Overcast','Rainy','Rainy','Rainy','Overcast','Sunny','Sunny','Rainy',
'Sunny','Overcast','Overcast','Rainy']
```

[2]:
```
#Second Feature
temp =␣
↪['Hot','Hot','Hot','Mild','Cool','Cool','Cool','Mild','Cool','Mild','Mild','Mild','Hot',
```

[3]:
```
# Label or Target Variable
play = ['No','No','Yes','Yes','Yes','No','Yes','No','Yes','Yes','Yes','Yes',
'Yes','No' ]
```

[4]:
```
# import Label Encoder
from sklearn import preprocessing
```

[5]:
```
#creating Label Encoder
le = preprocessing.LabelEncoder()
```

[6]:
```
#converting Strigns Labels to numbers
weather_encoded = le.fit_transform(weather)
```

[7]:
```
weather_encoded
```

[7]:
```
array([2, 2, 0, 1, 1, 1, 0, 2, 2, 1, 2, 0, 0, 1])
```

[8]:
```
#converting Strigns Labels to numbers
temp_encoded = le.fit_transform(temp)
label_col = le.fit_transform(play)
temp_encoded
```

[8]:
```
array([1, 1, 1, 2, 0, 0, 0, 2, 0, 2, 2, 2, 1, 2])
```

[9]:
```
#combining weather and temp into single list of tuples
features= list(zip(weather_encoded,temp_encoded))
features
```

[9]:
```
[(np.int64(2), np.int64(1)),
 (np.int64(2), np.int64(1)),
 (np.int64(0), np.int64(1)),
 (np.int64(1), np.int64(2)),
 (np.int64(1), np.int64(0)),
 (np.int64(1), np.int64(0)),
 (np.int64(0), np.int64(0)),
 (np.int64(2), np.int64(2)),
 (np.int64(2), np.int64(0)),
 (np.int64(1), np.int64(2)),
 (np.int64(2), np.int64(2)),
 (np.int64(0), np.int64(2)),
 (np.int64(0), np.int64(1)),
 (np.int64(1), np.int64(2))]
```

[10]:
```
from sklearn.neighbors import KNeighborsClassifier
model = KNeighborsClassifier (n_neighbors=3)
model.fit(features,label_col)
KNeighborsClassifier(n_neighbors=3)
```

[10]:
```
KNeighborsClassifier(n_neighbors=3)
```

[11]:
```
#predict output
predicted = model.predict([[1,0]])
predicted
```

[11]:
```
array([1])
```

[12]:
```
print(predicted)
```

```
[1]
```